

400 Hz Converter STATEMENT OF WORK (SOW)

1.0 BACKGROUND

The Naval Surface Warfare Center, Dahlgren Division (NSWCDD) requires one (1) Series SFC, 100kVA, Static Frequency Converter with installation services to establish a fully redundant 400Hz power system for the [REDACTED] laboratory, ensuring operational continuity for mission-critical test and evaluation.

2.0 APPLICABLE DOCUMENTS

- Power Systems & Controls (PS&C) Proposal MQ26-0040
- Original Equipment Manufacturer (OEM) Specifications
- National Electrical Code (NEC)

3.0 REQUIREMENTS

3.1 The contractor shall provide one (1) each Series SFC, 100kVA, Static Frequency Converter, Model Number SFCD00100NLJ, manufactured by Power Systems & Controls (PS&C).

- Rating: 100kVA, 80kW
- Input Voltage: 480Y/277 Volts, 3 Phase, 4W+G, 60Hz, Wye
- Output Voltage: 115 Volts, 3 Phase, 400Hz, Ungrounded, Delta 3 Wire
- Enclosure: NEMA 1 (Indoor Enclosure)
- Footprint: 44"W x 63"H x 32"D, Weight Approx. 2100 lbs 3.2

3.2 The contractor shall ensure the model meets the following specifications as detailed in the proposal:

- 12 Pulse PWM Solid-State Technology
- Double Conversion Topology & Galvanic Electrical Isolation
- Output Frequency Regulation: 400Hz (+/- 0.4Hz)
- Output Voltage Regulation: +/- 1% (Linear load)
- Overload Capability: 110% Continuous, 125% for 10 Minutes
- Protections: Over/Under Voltage, Short Circuit, and Overload with Auto Shutdown
- Indicators: LCD Real-Time Status/Clock and Self-Diagnosis LEDs

3.3 The contractor shall deliver the SFC FOB to the NSWCDD, located in Dahlgren, Virginia.

3.4 The contractor shall provide all labor, tools, and materials to install the new SFC to create a redundant power configuration alongside the existing converter. All work, including installation, comprehensive testing, and coordination with the existing system, must be fully completed within a strict five (5) consecutive day period. The configuration will allow only one converter to support the load-bus at a time via a manual transfer switch. The contractor shall:

3.4.1 Provide and install a 225A 480V three-phase MCB distribution panel (or equivalent) to split the existing source power upstream of the two converters.

3.4.2 Provide and install a 600A three-pole manual transfer safety switch (or equivalent) for 400Hz application to switch the load between the existing and new converters.

3.4.3 Mount new panels and switching components on the same wall as the converters using blind-type fasteners.

3.4.4 Provide and install all necessary installation material, including new appropriately sized conductors, conduit, and/or raceways from the new source panel to the new converter and from the new converter to the new transfer switch, per NEC requirements.

3.4.5 Make connections to the existing source and load conductors using mechanical splice components as needed. Splice connections may be located under the raised floor.

3.4.6 Upon completion of the physical installation, perform startup services to energize and test the system. This testing must encompass both the new system independently and its coordination with the existing legacy system. The contractor must verify that manual failover and redundant operations function seamlessly and within OEM tolerances before the 5-day completion window expires.

3.5 Warranty

The contractor shall provide a minimum of a one (1) year standard warranty for the Static Frequency Converter, which includes coverage for all parts and on-site service.

4.0 GOVERNMENT-FURNISHED PROPERTY/MATERIAL/INFORMATION (GFP/GFM/GFI)

4.1 The government will perform the following prior to scheduling the installation:

4.1.1 Provide a structure, providing both support and a minimum of 6 inches of rear clearance.

4.1.2 Verify the existing cables are in good condition and capable of being bottom fed through the front of the SFC.

4.1.3 Upon receipt of the SFC noted above, unload and position the SFC on the structure.

4.2 The government will coordinate with the contractor to schedule installation services once acceptance of the equipment has occurred.

4.3 The government will coordinate access to the NSWCDD location to complete the work.

5.0 TRAVEL

Travel for three (3) technicians to Dahlgren, Virginia, is required. The entire on-site performance period for installation, system coordination, and final startup testing shall not exceed **five (5) days**. All

travel and scheduling shall be strictly coordinated with the technical point of contact to ensure facility availability for this uninterrupted 5-day block.

6.0 TECHNICAL POINT OF CONTACT

